

A Controlled Comparison of Model Architectures and Target Transformations for Predicting ESA CCI Biomass from Fused Sentinel-1/Sentinel-2 Imagery

Sarbani Das, Silvia Pepino

Affiliation

Globeeye

sarbani@globeeye.eu, silvia@globeeye.eu

Abstract

Global above-ground biomass (AGB) products such as ESA’s Climate Change Initiative (CCI) Biomass provide wall-to-wall coverage at 100 m resolution. We compare a residual U-Net against a per-pixel Random Forest for predicting ESA CCI Biomass from Sentinel-1 and Sentinel-2 imagery over a 77 by 51 km block of the central Amazon, and cross both with a raw and a log-transformed target. All four models see the same tiles, labels, split and seed. The log transform hurts both model families, adding 6.7 Mg/ha of underprediction to the network and 10.8 to the forest and raising RMSE in each case, which places the cause in the back-transform rather than in the architecture. The biomass in this block is in fact mildly left-skewed, so the usual reason for applying the transform does not hold here. Once the baseline shares the transform, the architecture gap is 1.8 Mg/ha RMSE, and the network wins on only 40 of the 76 test tiles, with a median per-tile advantage of 0.15 Mg/ha. Finally, 10.9% of label cells carry CCI’s zero for non-forest. Including them raises R^2 from 0.15 to 0.72, and on forested cells alone all four models are worse than a constant predictor above 250 Mg/ha, where 57% of the cells lie. All figures are in Mg/ha.

1 Introduction

Above-ground biomass (AGB) underpins terrestrial carbon accounting, and the Amazon is where the stakes are highest. Field plots and airborne LiDAR give the most trustworthy numbers but cannot cover a basin, so wall-to-wall products such as ESA CCI Biomass fill the gap by trading resolution for coverage. At 100 m, CCI cannot resolve selective logging or forest edges, which is the usual motivation for learning a finer-grained mapping from Sentinel imagery.

Predicting AGB from Sentinel-1 and Sentinel-2 with a CNN is by now well-trodden ground, and we make no claim to a new architecture. Our interest is in three things that such comparisons routinely leave entangled. First, when a network is reported to beat a per-pixel baseline, the two often differ in more than the architecture, and in our own first configuration the baseline carried a log-transformed target while the network did not. Second, a network with a wide receptive field is scored against labels that are constant within each 100 m cell, so part of any advantage is smoothing rather than retrieval. Third, a reference product contains values that are easy to predict and are not biomass at all. We ran a two-by-two over architecture and target transform with everything else held fixed, and evaluated it so that these three effects come apart.

2 Related Work

U-Net-style encoder-decoder networks [3] with residual blocks [2] are the standard architecture for continuous retrieval from Earth Observation imagery, canopy height and AGB among them, and Random Forest regression on per-pixel spectral and backscatter features remains the usual baseline. We combine both in the conventional way. What we contribute is the design of the comparison and the accounting of what the reference product supplies.

3 Method

Table 1: Test-set performance in Mg/ha. Left, the primary evaluation over forested cells, after averaging predictions within each 100 m cell. Right, the same models with the 17.19% of non-forest cells included at their CCI value of zero, per 10 m pixel. The mean-value row predicts each subset’s mean everywhere, so its RMSE is that subset’s label standard deviation. Bias does not change under the spatial averaging.

Model	Target	forested cells, 100 m				all cells, 10 m		
		RMSE	MAE	R^2	Bias	RMSE	R^2	Bias
Mean-value	–	72.66	–	0.000	0.00	118.14	0.000	0.00
U-Net	raw	66.92	53.91	0.152	–19.54	62.88	0.717	–12.40
Random Forest	raw	68.71	55.10	0.106	–7.69	65.16	0.696	–4.79
U-Net	log1p	71.57	57.08	0.030	–23.53	67.33	0.675	–19.13
Random Forest	log1p	70.64	56.64	0.055	–19.23	67.11	0.677	–15.56

to non-forest, so these are not necessarily missing data. Querying the product’s own validity mask over all 380 tiles resolves it: 94.6% of the zero cells are unmasked and are therefore recorded measurements of no woody biomass, and genuinely masked pixels amount to 0.62% of the total. We keep them in training, since they are valid observations, but report metrics both over all cells and over forested cells alone, defined as those with a non-zero label. The forested subset is the primary evaluation.

Tiles were split 80/20 into 304 training and 76 test tiles in index order, which puts the test set in the southern strip. That strip includes the whole southern edge row, so its zero fraction is 17.19% against 9.38% and its label standard deviation is higher (118.14 against 100.90 Mg/ha with the zeros in). That gap is land cover, not forest structure.

3.2 Models, target and training

The network is a U-Net encoder–decoder with residual blocks and skip connections: four downsampling stages ($64 \rightarrow 128 \rightarrow 256 \rightarrow 512$, each a residual block with 2×2 max-pooling and Dropout2d at 0.25), a bottleneck of two residual blocks at 1024 channels, four upsampling stages with transposed convolutions and skip concatenation, and a 1×1 convolution to one output channel. The baseline is a Random Forest, 200 trees at depth 15, fitted to per-pixel vectors of the same eight bands with the same labels and split, subsampling 2000 pixels per tile. It predicts every test pixel rather than a subsample, so its R^2 shares a denominator with the network’s.

A log transform is the standard response to a right-skewed biomass target, on the argument that rare high-biomass cells carry little weight under squared error. That argument does not apply here. Over the forested test cells the label has a mean of 259.5 and a median of 263.0 Mg/ha with a skewness of -0.35 : the distribution is mildly left-skewed and its mass sits high, 57% of cells above 250 Mg/ha and 0.8% below 50. We include it anyway, since the assumption is applied routinely and is worth testing. Both models are fitted twice, to z-score-normalized AGB and to z-score-normalized $\log(1 + \text{AGB})$, the latter returned to Mg/ha by undoing the z-scoring and applying $\exp(\cdot) - 1$, with predictions clipped at zero in both branches. The networks use Adam at 1×10^{-3} , batch size 4, 15 epochs and seed 42, with no held-out validation set and no checkpoint selection.

4 Results

Table 1 gives the four models on the 76 held-out tiles, which hold 77,824 CCI cells, 64,445 of them forested.

The log transform hurts both model families. It costs the U-Net 4.65 Mg/ha of RMSE and the Random Forest 1.93 on forested cells, and adds 6.73 and 10.77 Mg/ha of underprediction respectively. A Random Forest has no gradient training and no receptive

Table 2: RMSE in Mg/ha within bins of the reference value, on forested 100 m cells. The mean-value column is a constant predictor at the global mean of 259.5 Mg/ha, which is strong near the mode by construction. Bins are defined on the reference value, so these are not calibration curves.

AGB bin	n	Mean-value	U-Net raw	RF raw	U-Net log1p
0–50	502	233.7	65.3	114.5	50.3
50–100	957	181.9	104.2	134.6	98.9
100–150	2,954	131.0	97.2	115.9	98.6
150–200	8,111	82.9	62.0	75.7	64.2
200–250	14,936	36.2	27.5	34.6	30.8
250–300	17,681	20.6	37.5	28.1	42.0
300+	19,304	87.5	96.8	91.7	104.8

field, so an effect that appears in both cannot be a property of the network.

The architecture gap is small, and unevenly distributed. At a matched raw target the U-Net leads by 1.79 Mg/ha RMSE and 0.046 R^2 on forested cells; at the log target the Random Forest is ahead on both. Per tile the picture is weaker still. The U-Net has the lower RMSE on 40 of 76 tiles, the median advantage is 0.15 Mg/ha, and the mean of 2.88 is carried by a handful of tiles reaching 54.7. The advantage does not track tile label variance ($r = 0.04$) but does track the fraction of cells that are forested ($r = -0.35$), and it is 4.92 Mg/ha on the 22 tiles at the grid boundary against 2.05 on the 54 interior ones. What the network gains, it gains where labels are sparse.

Non-forest cells carry the headline number. Including them lifts the best R^2 from 0.152 to 0.717. Both terms move: the zeros raise the label standard deviation from 72.66 to 118.14 Mg/ha, and telling forest from water and cleared ground is an easy problem for C-band backscatter and the SWIR bands. On forested cells the best model improves on a constant predictor by 7.9% of RMSE.

Skill is confined to the sparse low-biomass tail. Table 2 splits the error by reference value. Every model beats the constant predictor below 250 Mg/ha, by a wide margin in the lowest bins. Above 250 every model loses to it, and those two bins hold 36,985 of the 64,445 forested cells, or 57%. The aggregate R^2 of 0.15 is an average of large gains on a fifth of the cells and losses on the majority. In the top bin the U-Net’s bias is -90.4 Mg/ha, close to the -91 that a constant predictor would incur, which is what collapse toward the mean looks like.

5 Discussion and limitations

Why the transform costs accuracy. Fitting in log space and inverting with $\exp(\cdot) - 1$ estimates the conditional mean with a downward bias. Jensen’s inequality gives $\mathbb{E}[\exp(X)] \geq \exp(\mathbb{E}[X])$ for non-degenerate X , which holds whatever the direction of skew, and Duan’s smearing estimator [1] is the standard correction. The two-by-two is what makes the attribution possible: both families move the same way, and the one with no gradient training moves further. The transform was also chosen on an assumption this block does not satisfy, which is worth stating because the assumption is rarely checked.

How much of the network’s advantage is smoothing. The U-Net sees a 320×320 neighborhood, the Random Forest one pixel, and the labels are constant within 10×10 blocks. Within-block error therefore carries no label information, and smoothing suppresses it. Averaging both models’ predictions to the 100 m grid before scoring removes that compo-

nent and shrinks the raw-target gap from 2.80 Mg/ha to 1.79, and the R^2 gap from 0.074 to 0.046. About a third of the measured advantage is smoothing. The per-tile results suggest most of the remainder is tied to the grid boundary rather than to biomass.

All four models underpredict, and the trees least of all. At a matched target the Random Forest is the less biased (-7.69 against -19.54 on forested cells), which is the opposite of what limited extrapolation from leaf averages would predict. The likelier account is the bimodal target: 9.38% of training cells sit at zero while the forested mode is near 260, and a model fitted to the mixture lands below the forested mode. Consistent with this, the bias grows for all four models when non-forest cells leave the metric but not the training set. Fitting the two land covers separately would test it.

What the evaluation can and cannot show. The reference is a modeled product that takes Sentinel-1 as an input alongside two of our eight channels, so agreement is partly circular. CCI also saturates at high biomass, which is where most of these cells lie and where all four models fail, and we cannot separate saturation in the reference from failure in the models. The 64,445 forested cells come from 76 contiguous tiles and are strongly autocorrelated, so the effective sample is far smaller than the count. Each configuration is a single seeded run, so the 1.79 Mg/ha gap carries no variability estimate; the transform effect, consistent across two model families, is firmer. Results describe one block of central Amazon terra firme forest.

6 Conclusion

Crossing a residual U-Net and a Random Forest with a raw and a log-transformed target, under identical data and seed, separates three effects that are usually reported as one. The transform degrades both architectures and adds underprediction to both, so the cause is the back-transform and not the network; it was also applied on an assumption of right skew that this block does not satisfy. The architecture gap is 1.79 Mg/ha at a matched target, a third of it smoothing on a replicated label field, and the network wins on barely half the test tiles. And the reference product’s non-forest zeros, 10.9% of cells, lift R^2 from 0.152 to 0.717 on their own, while on forested cells no model beats a constant predictor across the 57% of the range where most cells lie. Work worth doing next is separate treatment of forest and non-forest, a Random Forest given neighborhood features to isolate spatial context, validation against GEDI L4A footprints, and repeated seeds.

References

- [1] Naihua Duan. Smearing estimate: a nonparametric retransformation method. *Journal of the American Statistical Association*, 78(383):605–610, 1983.
- [2] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *CVPR*, pages 770–778, 2016.
- [3] Olaf Ronneberger, Philipp Fischer, and Thomas Brox. U-Net: convolutional networks for biomedical image segmentation. In *MICCAI*, pages 234–241, 2015.

A Data preparation and reproducibility

Sentinel-2 scenes were filtered to CLOUDY_PIXEL_PERCENTAGE below 80 and masked with the scene classification layer, excluding classes 0, 1, 3, 8, 9, 10 and 11, then reduced to an annual median composite for 2021. The eight input channels are z-scored per band using statistics from the 304 training tiles only, and label statistics come from the same tiles, in log space when the log target is in use. Non-forest cells carry a valid label of zero and entered both the label statistics and the loss.

All four models read one cached copy of the cleaned tile arrays. The forward and inverse target transforms, the label statistics and the metric function are shared code. Each model writes its test predictions to one file and every number reported here is computed from that file, so no two metrics for a model can come from different prediction sets.

Metrics at 10 m and at 100 m differ because averaging within a block removes within-block prediction variance; bias is unaffected. On forested cells at 10 m the figures are 68.04 (R^2 0.123) and 70.84 (0.050) for the raw-target U-Net and Random Forest, and 73.78 (-0.031) and 73.28 (-0.017) for the log-target pair.